

Comparing Analog CMOS Hopfield and Ising Networks on NP-hard Problems

Hopfield vs Ising on the SoC FPAA

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Abstract—Analog circuits can be used for physical energy minimization, efficiently solving combinatorial optimization problems and having the potential to achieve speedups over digital solvers. This paper presents a direct comparison of two analog optimization networks: Hopfield and Ising. Both networks are compared on a large-scale Field Programmable Analog Array (FPAA) by solving two NP-hard problems (max-cut and associative memory) over various initial conditions. The energy, convergence time, power, and system complexity of the two networks is compared where it is found that Hopfield networks outperformed the Ising network in the experimental test cases.

Keywords—Hopfield, Ising, FPAA

I. ENERGY SURFACE MINIMIZATION

NP-HARD problems are computationally intractable in digital systems; exact solutions run in exponential time and more practical polynomial solvers often find suboptimal solutions ([1], [2]). Energy surface minimization is a technique used in practical solvers where an energy surface is constructed such that its minima are at the optimal solutions; finding the minimum of such a surface then finds the solution to the problem [3]. Effective uses of this technique require both the correct energy surface and minimization framework.

Physical systems can solve energy minimization problems, with early examples connecting to Ising's models of magnetism [4], that then inspired Hopfield networks ([5]–[7]) which were in turn the foundation of many Ising network implementations ([8]–[10]). When the energy surface of NP-hard problems are put onto these systems they minimize, finding optimal solutions some of the time and good solutions most to all of the time, making them very interesting for practical applications.

Analog circuits have been used as a physical medium for both Hopfield and Ising networks ([7], [9], [11], [12]). Ising networks consist of a fully connected system of coupled oscillators, where the energy surface can be described as

$$\mathbf{H}(\mathbf{s}) = -\mathbf{s}^T \mathbf{J} \mathbf{s} - \mathbf{h}^T \mathbf{s}, \quad (1)$$

where $\mathbf{s}$ is a vector describing the spin of each state (± 1 for a binary network), $\mathbf{J}$ is a weight matrix describing the

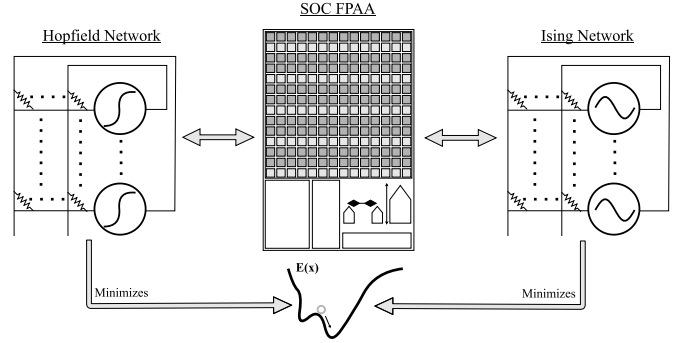


Fig. 1. Both Hopfield and Ising networks dynamically minimize energy surfaces, allowing for good or optimal solutions to otherwise computationally difficult problems. The question of which implementation has an advantage is tested on the SoC FPAA to give a level playing field and fair comparison.

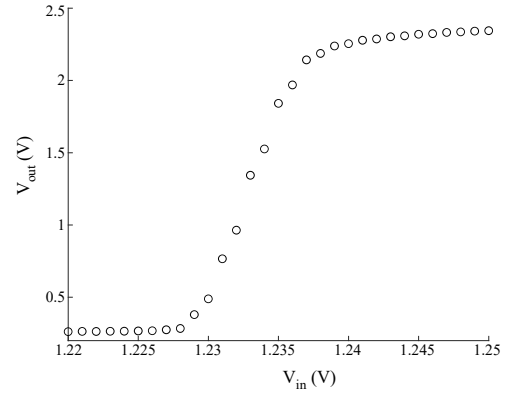
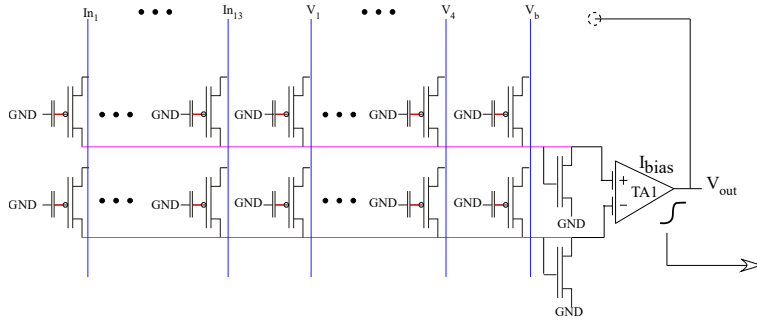
connections between oscillators, and $\mathbf{h}$ is a constant bias [13]. Hopfield networks have a very similar energy function

$$\mathbf{E}(\mathbf{x}) = -\frac{1}{2} \mathbf{x}^T \mathbf{W} \mathbf{x} - \theta^T \mathbf{x}, \quad (2)$$

where the state variable, $\mathbf{x}$ describes the output of a sigmoidal perceptron rather than an oscillator phase [5], and θ is a constant bias. Aside from the state variable, the two networks are constructed similarly (all-to-all connections) and scale identically with problem size, typically requiring $O(N)$ Hopfield/Ising nodes.

When choosing an analog energy minimization network, should a Hopfield or Ising network be chosen? This paper will provide insight into that question, focusing specifically on CMOS, analog circuits for Ising and Hopfield networks. The two networks will first be constructed on the large-scale Field Programmable Analog Array (FPAA) [14], allowing for direct comparison on the same silicon 350nm substrate (Fig. 1) (Section II). Then the same quadratic minimization problems will be solved on both (Section III), and their behavior compared and contrasted (Section IV).

Hopfield Neuron (Single CAB)



Ising Neuron (Single CAB)

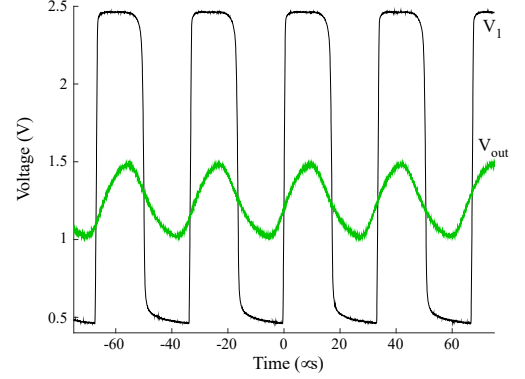
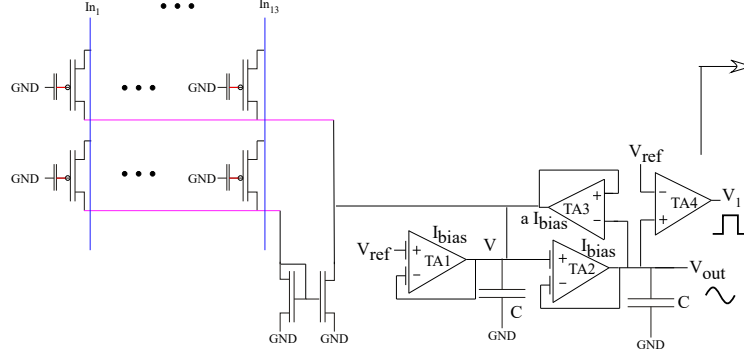


Fig. 2. Circuit diagrams of a single neuron in the Hopfield and Ising networks along with their corresponding outputs. Both neurons fit in one CAB of the FPAA, but the Ising neuron takes up the entire CAB while four Hopfield neurons can fit inside one CAB. The Hopfield network uses a single and two VMM rows. The Ising network uses a MeadSOS oscillator made of three TAs, two rows of a VMM with an Nmirror, and an external buffer to rail the signal.

II. ANALOG HOPFIELD AND ISING NETWORKS

A fair comparison between networks requires equitable devices and conditions of operation; comparing different technologies such as a 1 GHz nano-oscillator with a 1 μm CMOS process Transconductance Amplifier (TA) is hardly fair. In this work a reconfigurable analog computing platform (Section II-A) is used to build both Hopfield (Section II-B) and Ising networks (Section II-C) in 350 nm CMOS [15]. The same circuit elements are used on the same chip, leveling the technological playing field for our comparison.

A. Source-Coupled VMM

Vector Matrix Multiply (VMM) crossbars are automatically present in the routing fabric of the reconfigurable SoC FPAA. Because the switch elements of the FPAA are also precisely programmable VMMs can be built out of the routing fabric rather than requiring specialized blocks. Input voltages come to the source of a FG pFET and weigh the resulting current, which can be modeled in subthreshold as

$$\begin{aligned} V_s &= V_{s0} + \Delta V_s \\ V_d &= V_{d0} + \Delta V_d \\ I &= I_{s0} W e^{(\Delta V_s - \sigma \Delta V_d) / U_T}, \end{aligned} \quad (3)$$

where W is a positive weight that can be programmed by changing the V_Q on the FG node. When multiple weighted FG pFETs are connected, KCL adds the weighted currents

up to create a VMM. Both Hopfield and Ising network formulations have weight matrices that define the coupling between neurons, and in both cases a source coupled VMM implements this behavior on the FPAA.

B. Hopfield Network

The weighted connections into a Hopfield neuron result in an output along a sharp sigmoid

$$S_i = \begin{cases} 1, & \sum_j O_{ji} \geq T \\ 0, & \sum_j O_{ji} < T \end{cases}, \quad (4)$$

modeled on the FPAA with a TA (Fig. 2b) that outputs a sigmoidal voltage according to the difference in voltage on its positive and negative terminals [12]. This allows for signed weights, where positive and negative weights go to the corresponding VMM row for the neuron. A transistor to ground is used to convert the weighted current on each row to a voltage that the TA can then compare.

C. Ising Network

The oscillator for the Ising network on the FPAA is generated by a Hopf bifurcation circuit: the MeadSOS [16], which has known oscillatory dynamics created from the non-linear feedback element. By putting an oscillatory current into the feedback node, coupling between oscillators can occur. This coupling signal can be weighted by a VMM, allowing for Ising

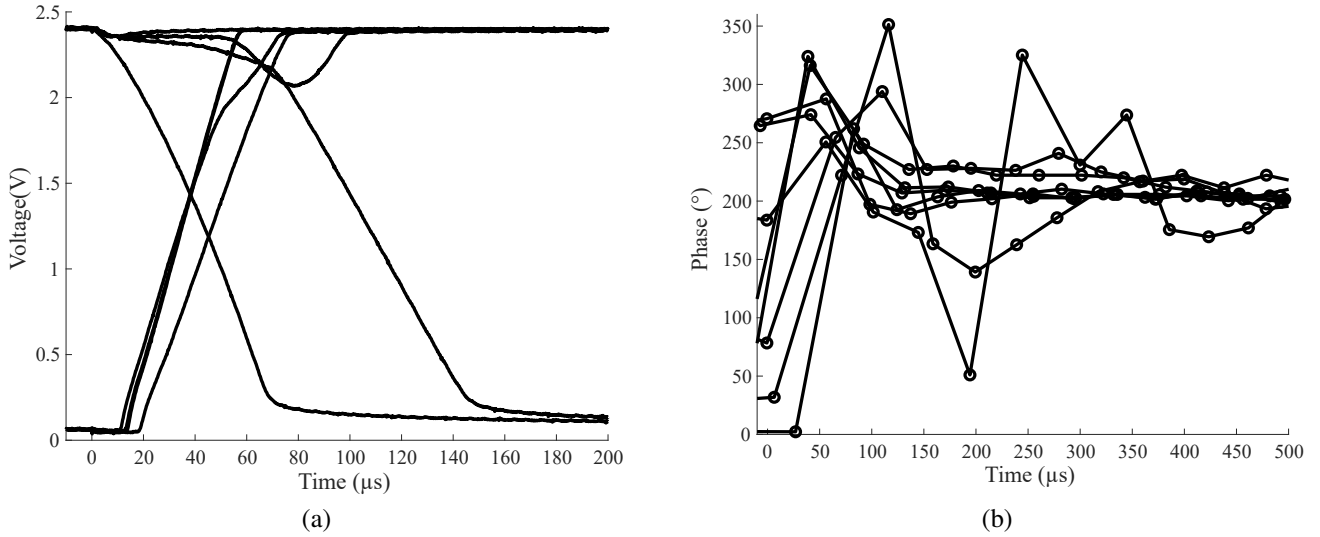


Fig. 3. a) Single neuron of the Hopfield network converging to its final state over eight different initial conditions. An initial condition is forced through an external static input, and then relaxed to allow convergence. b) Single neuron of the Ising network converging to its final state over eight different initial conditions. An initial condition is set by uncoupling the neurons from each other. The network is then instantly coupled together to allow for convergence.

networks to be formed out of coupled MeadSOS oscillators. A signed weight can be achieved by using two rows of a VMM similar to the Hopfield network where one row adds current into the feedback node and the other goes into an nFET mirror which takes current away from the node (Fig. 2c). Current going into the feedback node corresponds to a positive weight (matched phases) while current leaving the node corresponds to a negative weight (opposite phases).

III. SOLVING NP-HARD PROBLEMS

Both Hopfield and Ising networks minimize quadratic energy surfaces as described in (2) and (1). These energy expressions can be written in summation form as well

$$E(\mathbf{x}) = -\frac{1}{2} \sum_i^N \sum_j^N W_{ij} x_i x_j - \sum_k^N I_k x_i, \quad (5)$$

for Hopfield ([7]) and

$$H(\mathbf{s}) = -\sum_i^N \sum_j^N J_{ij} s_i s_j - \sum_k^N h_k s_i, \quad (6)$$

for Ising [13]. If the ground state of these energy functions describes the solution to a problem, Hopfield and Ising networks can find good to optimal solutions to otherwise computationally difficult problems. The NP-hard class of problems are one example of such problems, and quadratic energy surfaces that describe these problems have been formulated [13].

One can transform a problem mapping between Hopfield and Ising networks with a state variable substitution

$$s_i = 2x_i - 1, \quad (7)$$

allowing for both networks to solve the same problem. This procedure was performed for the max-cut and associative memory problems.

The max-cut problem can be described as

$$H = B \sum_{(uv) \in E} \frac{1 - s_u s_v}{2} \quad (8)$$

for an Ising network, where

$$W_{ij} = \begin{cases} B * w_{ij} & i, j \in E \\ 0 & \text{otherwise,} \end{cases} \quad (9)$$

$$h_k = 0, \quad (10)$$

and w_{ij} is the edge weight between vertices i and j ([17]). Using the substitution for Hopfield networks (Eqn. 7) gives

$$W_{ij} = \begin{cases} B * w_{ij} & i, j \in E \\ 0 & \text{otherwise,} \end{cases} \quad (11)$$

$$I_k = -\frac{1}{2} \sum_j W_{kj}, \quad (12)$$

where w_{ij} is the same as in the Ising case.

This procedure can be repeated for the Associative Memory (AM) problem, where the Hopfield energy function is described as

$$W_{ij} = \sum (2V_i - 1)(2V_j - 1) \quad (13)$$

$$I_k = 0, \quad (14)$$

where each neuron represents one bit in the stored pattern ([5]). Substituting (Eqn. 7) for Ising results in

$$J_{ij} = \sum s_i s_j \quad (15)$$

$$h_k = 0, \quad (16)$$

where in both cases the weight matrix is

$$W_{ij} = J_{ij} = \begin{cases} 1 & V_i = V_j \\ -1 & V_i \neq V_j, \end{cases} \quad (17)$$

resulting in a full all-to-all network.

Size	Problem	Hopfield Convergence Time (μ s)					Ising Convergence Time (μ s)				
		N1	N2	N3	N4	Total Avg	N1	N2	N3	N4	Total Avg
3	MaxCut 2E	33.68	63.62	47.84	–	48.38	0	485.44	710.24	–	398.56
	MaxCut 3E	63.75	92.80	44.83	–	67.13	0	940.16	675.56	–	538.57
	AM 001	41.80	69.36	55.65	–	55.60	0	817.36	469.28	–	428.88
	AM 111	16.41	19.03	18.76	–	18.07	0	391.56	288.44	–	226.67
4	MaxCut 3E	146.10	145.54	133.51	120.76	136.48	0	269.92	332.92	715.6	329.61
	MaxCut 4E	92.37	96.57	92.88	107.02	97.21	0	339.44	826.24	578.48	436.04
	MaxCut 5E	203.90	120.53	76.18	176.28	144.22	0	250.84	208.36	212.64	167.96
	MaxCut 6E	71.22	102.70	56.76	96.68	81.84	0	617.88	392.96	788.48	449.83
	AM 0001	108.88	86.97	115.36	147.45	114.66	0	168.68	222.96	358.28	187.48
	AM 1100	166.60	191.04	157.77	142.79	164.55	0	410.16	244.12	191.12	211.35
	AM 1111	116.89	68.32	80.11	60.80	81.53	0	172.32	46.96	215.28	108.64
Averages		96.51	96.04	79.97	121.68	91.79	0	443.16	401.64	437.13	316.69

Fig. 4. Average convergence time in three and four neuron Hopfield and Ising networks solving the max-cut and associative memory problems. Each network is started from 8 different initial conditions, and the convergence of each neuron is measured individually. The binary max-cut problem is labeled by the number of edges, or E. The associative memory (AM) problem is labeled by the binary pattern stored. On average the Hopfield network converges faster than the Ising network.

Size	Problem	Hopfield Convergence Energy (pJ)					Ising Convergence Energy (pJ)				
		N1	N2	N3	N4	Total Avg	N1	N2	N3	N4	Total Avg
3	MaxCut 2E	0.15	0.06	0.06	–	0.09	0	472.09	690.74	–	387.61
	MaxCut 3E	0.09	0.10	0.12	–	0.10	0	926.22	665.59	–	530.60
	AM 001	0.24	0.15	0.15	–	0.18	0	805.18	462.28	–	422.48
	AM 111	0.01	0.06	0.14	–	0.07	0	385.71	284.15	–	223.29
4	MaxCut 3E	3.95	0.41	0.65	1.08	1.52	0	272.88	328.04	705.01	326.48
	MaxCut 4E	0.22	0.40	0.86	0.46	0.49	0	342.92	834.72	584.25	440.47
	MaxCut 5E	1.98	1.67	2.41	1.29	1.84	0	253.44	210.54	220.13	171.03
	MaxCut 6E	0.30	0.46	0.02	0.62	0.35	0	639.86	406.92	816.37	465.79
	AM 0001	6.62	5.95	7.57	0.03	5.04	0	174.83	231.09	370.89	194.20
	AM 0011	3.43	1.76	6.76	6.11	4.52	0	424.89	252.91	197.79	218.90
	AM 1111	6.89	3.94	4.23	3.51	4.64	0	178.52	48.69	223.18	112.59
Averages		2.17	1.36	2.09	1.87	1.71	0	443.32	401.42	445.37	317.59

Fig. 5. Averaged, approximate energy consumption of a three and four neuron Hopfield and Ising network solving the max-cut and associative memory problems. Each neuron was started from 8 different initial conditions and allowed to converge. The energy consumption of the buffers on both networks was excluded, as the numbers would be very similar and make a direct comparison less clear. On average the Hopfield network consumes less energy than the Ising network.

IV. COMPARISON OF HOPFIELD AND ISING NETWORKS

The state variable difference between Hopfield and Ising networks, while mathematically minor, results in tangible differences between the two networks in metrics such as power, energy, and convergence time even when both networks can correctly solve identical problems. To test these differences both networks were programmed to solve different max-cut and AM problems (Section IV-A) and evaluated over their convergence period (Section IV-B). Using these results we can extrapolate to future efficient analog optimization networks (Section IV-C).

A. Problem Setup

The initial condition was set on a Hopfield network through the activation of external inputs that forced a non-ideal state. Because of the high gain sigmoidal nonlinearity most states of the Hopfield network are binary, so the initial conditions

also forced a binary unstable starting point. The external input was then released, allowing the network to converge back to a stable energy minimum (3a).

On the Ising network a random initial condition was set by setting V_{ref} to Gnd, uncoupling the network. The neuron phases were allowed to drift apart, and then V_{ref} was set back to the midpoint of the oscillations, around 1.3 V, which coupled the oscillators together and caused convergence to a final state (3b). An external signal was applied at twice the frequency of the oscillators to force convergence to two different states. Because neuron 1 is always used as a reference, it always converges in zero time and consumes zero energy.

B. Comparing Energy and Convergence Time

Both networks converged to the optimal solution from every trial in both case studies, regardless of initial condition. The time taken to converge to this optimal final solution depends on the type of network, the initial conditions, and the programmed

problem. Over the eight initial conditions for each problem, the Hopfield network consistently converged faster than the Ising network (Fig. 4), averaging around 100 μ s overall while the Ising network averaged 425 μ s.

The energy taken by the networks over their convergence window is intrinsically tied to the convergence time. Energy calculations agree with this (Fig. 5), with the Ising network requiring more energy than the Hopfield network. Other work also follows this trend [8], [18]. The energy consumed by the buffers in both networks was omitted as they consume a similar amount of energy and are not intrinsic parts of the network structure. In addition, the energy provided by external inputs, such as the subharmonic injection locking (SHIL) signal for the Ising network, was omitted.

C. Efficient Analog Minimization Networks

These results suggest that Hopfield networks are the superior CMOS implementation of efficient energy minimization networks. We expect these results to hold through equitable technological scaling because the core feature, static voltage level versus oscillation, remains the same independent of technology node. Ising networks have been shown to benefit from advanced oscillators, but any advancements in technology could also benefit the Hopfield network.

The FG devices used on the FPAA for comparison would be favorable in a custom Hopfield implementation. FG devices show 14-bit programming precision and are one of the first computing-in-memory devices [19]. Crossbars of $m \times n$ FG devices have been demonstrated at an area of $6.5(m/4) \times 18.325(n/2) \mu\text{m}^2$ in 65nm CMOS [20], allowing for a fully connected 50x50 crossbar of size $81.25 \times 458.125 \mu\text{m}^2$, smaller than the (estimated) $425 \times 650 \mu\text{m}^2$ Ising network presented in [11] (neglecting the small area of external programming circuitry and the sigmoidal perceptron). A locally connected architecture in 65nm would also have an estimated area similar to [21], but with the high weight programmability that FG devices provide and potential energy increases with the Hopfield topology. FG bandwidth estimates show a quadratic function of minimum process size, predicting 4 GHz speed in 40nm compared to 55 MHz in 350nm, allowing for speed increases as well [15]. Therefore a custom FG-based analog Hopfield network could lead to a highly dense, highly programmable, efficient energy minimization network.

Reconfigurable energy minimization schemes are another area that an FG based Hopfield network would excel, allowing for efficient use of space for more sparse problems but still keeping connectivity for denser problems. A proposed method for a reconfigurable Ising machine [22] uses an architecture similar to the Manhattan routing used in FPGAs and the FPAA. The SoC FPAA uses in-routing computation for maximum efficiency, and a reconfigurable platform allows for easy interfacing for custom end-to-end systems depending on application. Combining the proven efficiency and computational power of the SoC FPAA [15] with the numerical programming density of FG devices would lead to a powerful, reconfigurable, programmable Hopfield network.

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